TEJAS PATEL

Hands-on AI Architect & Technical Lead | Agentic AI • Intelligent Test Automation • Model Fine-Tuning

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PROFESSIONAL SUMMARY

Hands-on AI architect and engineer with **23 years** of experience. I design and build **agentic AI applications** myself, mostly in Python, R and TypeScript. At CDC I built a Playwright crawler that tests a **STARLIMS lab system** with no hand-written scripts. On my own time, under Algocrat Labs, I build AI products including a federal procurement intelligence platform, **Dragon Farm** (an R package for SLM fine-tuning) and a D3 data storytelling pipeline. Strong background in data architecture, HL7 healthcare data and federal health IT. I lead the automation team at CDC and build alongside them.

Agentic Systems ◆ **AI Test Automation** ◆ **Small Language Models** ◆ **Data Storytelling** ◆ **Federal Health IT**

TECHNICAL SKILLS

Agentic & GenAI: LangChain, LangGraph, Microsoft Agent Framework, Copilot Studio, Claude, Azure OpenAI, AWS Bedrock, RAG, GraphRAG, Ollama

ML & Fine-Tuning: PyTorch, TensorFlow, Hugging Face, LoRA / QLoRA, SLMs, ESM-2, model monitoring, drift detection, bias mitigation

Test Automation: Playwright, agentic crawlers, Page Object Model, GitHub Actions, CI/CD, Grafana

Cloud: AWS (SageMaker, Bedrock, Lambda, Batch, ECS, DynamoDB), Azure (OpenAI, Synapse, AKS, Event Hubs)

Data: PostgreSQL, SQL Server, MongoDB, Databricks, Spark, Kafka, Elasticsearch, Neo4j, Redis, DuckDB, HL7

Languages & Viz: Python, R, TypeScript, Java, Scala, C# • D3.js, React, Angular, Tableau, Power BI, Plotly

INDEPENDENT AI PROJECTS

Side projects I build on my own time under the Algocrat Labs name, separate from my Booz Allen work.

AwardLens AI: Federal Procurement Intelligence

awards-dashboard.pages.dev

- Building a platform that helps contractors find and track federal opportunities. A **two-tier router** sends narrow and SQL-style questions to fine-tuned small models and broader reasoning to Claude.
- Runs almost entirely on free cloud tiers (Cloudflare and Oracle Always Free) to keep hosting costs near zero.

Dragon Farm: Drag-and-Drop SLM Fine-Tuning for R

dragonfarm.dev

- Built an R package that lets RStudio users **fine-tune small language models** (LoRA/QLoRA) through an agentic, drag-and-drop pipeline, similar to what Azure AI Foundry offers.
- Gives **R analysts and statisticians** a way to customize models without learning Python or MLOps.

DeViz and Informance.ai: Agentic Data Storytelling

deviz.io • informance.ai

- Built an agentic pipeline that reads long documents like SEC 10-K filings, pulls the numbers into JSON and turns them into **interactive D3.js data stories**. Completed a full story for Costco's 2024 10-K.
- Informance.ai packages the same idea for investor relations teams at mid-cap public companies, built on Astro and Cloudflare Pages.

PROFESSIONAL EXPERIENCE

Booz Allen Hamilton | Data & AI Technology Leader

Apr 2014 - Present

Technical Lead • AI and Automation for CDC Office of Laboratory Systems and Response (OLSR)

Agentic STARLIMS Test Crawler

CDC • Booz Allen

- Built an AI crawler that tests a STARLIMS (ExtJS) lab system starting from just **an entry point and a JSON workflow payload**. No hand-written scripts or test cases needed.
- Gave the agent workflow data, the app's HTML and workflow definitions so it knows what to test. It runs on Playwright and captures several locators per element, since ExtJS pages change their DOM a lot.
- Moved the team from writing tests by hand to **agentic testing**, one piece at a time. Now extending it so tests can be written straight from specs using Microsoft Agent Framework.

- **Own delivery to CDC**: set technical direction for the automation team, coach engineers and manage the client relationship.
- Applied protein language models to public health genomics, benchmarking **ESM-2 embeddings** against profile HMMs for antimicrobial resistance gene detection.
- Assessed **multi-agent orchestration frameworks** for enterprise AI work, focusing on task breakdown and production scale.
- Share agentic automation practices with the Booz Allen Atlanta AI Community of Practice through live demos.

Earlier at Booz Allen • Enterprise Data Architecture, Federal Health

- Led the digitization of a **national ambulatory care survey**, designing a reusable CQRS-based survey platform and governed pipelines for structured and HL7 healthcare data.
- Set up standardized **XDR data exchange** with major EMR vendors and built real-time pipelines for mortality data streams.
- Built a high-performance Apache Solr and .NET search and analytics API.

MediaBrix | Business Intelligence Architect **Aug 2012 - Apr 2014**

- Built Random Forest models to predict ad behavior for targeting, along with SSIS/SSRS pipelines and SSAS OLAP cubes.

EyeWonder | Business Intelligence Architect **Nov 2010 - Aug 2012**

- Designed ad analytics warehouses, a Naive Bayes fraud detection system and near-real-time conversion pipelines.

Earlier Roles | Database Developer, Consultant, Senior BI Developer **2003 - 2010**

- Wrote Microsoft Press certification exam content, built sentiment analysis engines and OLAP warehouses, and worked on optimization problems in healthcare and supply chain.

CERTIFICATIONS & EDUCATION

- **AWS Certified Machine Learning - Specialty** (2025-2028)
- **AWS Certified Solutions Architect - Associate** (2025-2028)
- **Project Management Professional (PMP), PMI** (2024-2027)
- **M.S. Industrial Engineering**, Georgia Institute of Technology
- **B.S. Mechanical Engineering**, SRT Marathwada University, India